

CARBOON

Whole-Life Carbon Calculator

Methodology Guidebook

Estimation Methods, Data Sources, Standards Reference,
and Assumptions for All Dashboard Sections (A-N)

Version 1.0 | July 2026

EN 15978 Compliant | LEED v5 Decarbonisation Ready

1. Introduction

The CARBOON Whole-Life Carbon Calculator is a web-based tool for estimating the whole-life carbon emissions of buildings across all lifecycle stages defined in EN 15978:2011 (Sustainability of construction works -- Assessment of environmental performance of buildings).

The tool combines a Python calculation engine (`carbon_engine.py`) with an interactive HTML dashboard (`result_visualization.html`). It accepts building geometry and type parameters, estimates material quantities from a parametric BoQ model, and calculates embodied carbon, operational carbon, end-of-life impacts, and circular economy credits across the full EN 15978 module structure (A-D).

The dashboard presents results through 13 visualisation sections (A-M), including doughnut charts, bar charts, time-series projections, a 3D building form visualizer, and carbon offset metrics. It is designed to support LEED v5 decarbonisation requirements by providing transparent, auditable carbon accounting at every lifecycle stage.

2. Standards & Frameworks Reference

The calculator is built on the following standards and frameworks:

- > EN 15978:2011 -- Sustainability of construction works. Assessment of environmental performance of buildings. Calculation method.
- > EN 15804:2012+A2:2019 -- Environmental product declarations. Core rules for the product category of construction products.

- > RICS Whole Life Carbon Assessment for the Built Environment (2nd Edition, 2023).
- > LEED v5 -- BD+C: New Construction, Energy & Atmosphere and Materials & Resources credits (USGBC).
- > CIBSE Guide F -- Energy Efficiency in Buildings.
- > BSRIA Rules of Thumb -- Benchmarking for building services.
- > UK Woodland Carbon Code -- Carbon sequestration rates.
- > ICE (Inventory of Carbon & Energy) Database v3.0 -- Embodied carbon factors for construction materials.
- > IPCC AR5 -- Global Warming Potential (GWP100) factors for refrigerants.

3. Calculation Methodology -- Lifecycle Stages A-D

3.1 Module A1-A3: Product Stage (Embodied Carbon)

A1-A3 covers the cradle-to-gate embodied carbon of construction materials, including raw material extraction (A1), transport to manufacturing facility (A2), and manufacturing (A3).

Where a Bill of Quantities (BoQ) is provided in the input JSON (boqReport field), A1-A3 carbon is calculated directly from material quantities multiplied by their respective ICE database emission factors. The engine's embedded ICE dataset covers 40+ common construction materials, including concrete grades, structural steel, rebar, timber products, insulation, glazing, and finishing materials.

Where no BoQ is provided, the engine estimates material quantities using a parametric model based on building area, number of floors, structural type, envelope type, grid spacing, and window-to-wall ratio. Each material is assigned to a building element (substructure, superstructure, envelope, internal walls, roof, services) for element-level tracking.

Key assumptions for parametric BoQ estimation:

- > RC frame: 0.35 m3 concrete/m2 GFA + 80 kg rebar/m3 concrete for substructure; columns (0.5x0.5m), beams (0.4x0.6m) at grid intersections; slabs at 200mm.
- > Steel frame: 90 kg/m2 structural steel + composite deck slab.
- > Timber frame: 0.25 m3 CLT/m2 GFA for structure; 0.15 m3/m2 for envelope.
- > Masonry: 215mm blockwork for external walls; 100mm blockwork for internal partitions.
- > Curtain wall / glass envelope: 35 kg/m2 aluminium framing + 25 kg/m2 glazing.
- > Reinforcement ratio varies by element: 2% for substructure and superstructure beams/columns; 1% for slabs.
- > Material densities from ICE database: concrete 2400 kg/m3, steel 7850 kg/m3, CLT 480 kg/m3, etc.

3.2 Module A4: Transport to Site

A4 carbon is estimated using a default transport distance of 50 km for local materials (concrete, aggregates, bricks, blocks) and 300 km for manufactured materials (steel, glass, insulation, timber products). Transport emission factors are 0.12 kgCO2e/tkm for road freight (DEFRA 2023).

3.3 Module A5: Construction / Installation

A5 covers on-site construction activities. Estimated at 6.5% of A1-A3 embodied carbon for site energy consumption, waste processing, and temporary works. This aligns with BS EN 15978 default scenarios and RICS WLCA guidance for typical UK/European construction.

3.4 Module B1: In-Use Emissions (Refrigerant Leakage)

B1 covers fugitive emissions from HVAC refrigerant leakage. When actual refrigerant data is provided in the OC_breakdown, it is used directly. Otherwise, default leakage rates of 2%/year for split systems and 5%/year for VRF systems are applied, with GWP values from IPCC AR5 (e.g., R-410A = 2088 kgCO2e/kg).

3.5 Module B2-B5: Refurbishment, Repair, Replacement

B2-B5 covers maintenance, repair, replacement, and refurbishment over the building's service life. Where actual data is provided in the JSON (EC_breakdown.B2 through B5), it is used directly. Where these fields are missing or zero, the dashboard marks them as 'Estimated' and assumes no additional in-use embodied carbon beyond what is captured in the upfront A1-A5 values.

IMPORTANT -- B2-B5 Status

The current 2_merged.json file has B_structural = 0, B_service = 0, and B_refurbish = 613,300 kgCO₂e. This means B5 (refurbishment) has actual data but B2-B4 are zero (no data provided, treated as no in-use embodied emissions for those stages). The dashboard badge will show 'Actual' for B5 and 'Estimated' for B2, B3, B4 -- reflecting the partial data completeness.

3.6 Module B6: Operational Energy

B6 covers carbon emissions from building energy consumption (heating, cooling, lighting, equipment, lifts, etc.). Calculated as:

$$B6 \text{ (kgCO}_2\text{e)} = \text{EUI (kWh/m}^2\text{/yr)} \times \text{GFA (m}^2\text{)} \times \text{Grid EF (gCO}_2\text{e/kWh)} / 1000 \times \text{Building Lifetime (years)}$$

Where EUI (Energy Use Intensity) defaults:

- > Large Office: 311 kWh/m²/yr (BSRIA/CIBSE benchmark)
- > Hospital: 450 kWh/m²/yr
- > Apartment (Mid-Rise): 180 kWh/m²/yr
- > Primary School: 200 kWh/m²/yr
- > Warehouse: 130 kWh/m²/yr

Grid Emission Factors (EF) are embedded for 20 countries x 4 decarbonisation scenarios (Conservative, Moderate, Ambitious, Net Zero) with linear interpolation between 2020, 2030, 2040, and 2050 data points. The Conservative scenario assumes gradual grid improvement of approximately 0.5-1.5% per year depending on country.

Where B6 is split into electricity and fossil fuel components (B6_elec, B6_fossil), the engine preserves this distinction. The default electricity ratio is 75%.

3.7 Module B7: Operational Water

B7 covers carbon from water supply and treatment. Estimated using water intensity benchmarks from CIBSE Guide G / BSRIA:

- > Large Office: 0.65 m³/m²/yr
- > Hospital: 0.95 m³/m²/yr
- > Apartment: 0.55 m³/m²/yr
- > School: 0.45 m³/m²/yr
- > CO₂ intensity of water: 0.35 kgCO₂e/m³ (supply + treatment)

When B7 data is not provided in the input JSON, this stage is marked as 'Estimated' in the dashboard.

3.8 Module C1-C4: End of Life

C1: Deconstruction / Demolition -- Estimated at 4.5 kgCO₂e/m² GFA for mechanical demolition (diesel plant, average from ICE demolition studies).

C2: Transport to disposal -- 50 km haulage distance at 0.12 kgCO₂e/tkm.

C3: Waste processing -- Material-specific processing emissions based on ICE end-of-life factors. Concrete crushing = 4.5 kgCO₂e/t; steel sorting = 2.5 kgCO₂e/t; timber chipping = 12 kgCO₂e/t.

C4: Disposal -- Landfill emissions based on material-specific decomposition factors. Inert materials (concrete, aggregate) = negligible; timber = 0.58 kgCO₂e/kg (methane from anaerobic decomposition); plastics/insulation = 0.02-0.05 kgCO₂e/kg.

Where the input JSON provides C_breakdown (C2, C3, C4) with actual data, the dashboard marks C1 as 'Estimated' and C2-C4 as 'Actual'.

3.9 Module D: Beyond System Boundary (Benefits & Loads)

Module D captures net benefits from reuse, recycling, and energy recovery beyond the building's system boundary. Calculated as the sum of avoided burden from material recycling:

$$D \text{ (kgCO}_2\text{e)} = \text{SUM} [\text{mass}_i \times \text{recycling_rate}_i \times (-\text{avoided_burden}_i)]$$

Recycling rates from ICE database range from 0-95% depending on material. Steel rebar achieves 95% recycling; concrete 75% downcycling to aggregate; timber 40% energy recovery. D values are typically negative (carbon credit).

4. LEED v5 Integration

The CARBOON dashboard is designed to support LEED v5 certification by providing transparent, auditable carbon accounting that maps directly to LEED v5 credits across the Materials & Resources (MR) and Energy & Atmosphere (EA) categories. This section explains how each dashboard metric feeds into LEED v5 credit requirements and how the integrated evaluation tab (Section N) automates point scoring and documentation.

4.1 Dashboard-to-LEED Credit Mapping

The following table maps dashboard sections to the LEED v5 credits they support:

LEED Credit	Dashboard Section	How the Data Maps
MRc2	Sections A, F, G, J	Embodied carbon (A1-A5) by material and by building element. BoQ quantities x ICE carbon factors per m ² GFA. Section N auto-computes EC density and reduction % vs baseline.
EAc3	Sections B, I, K	Operational energy (B6), EUI, and 25-year projection with grid decarbonisation. Section N computes EUI reduction % vs ASHRAE 90.1 baseline and maps to EAc3 point thresholds.
MRp2	Sections C, D	End-of-life waste management (C1-C4) and circular economy credits (Module D). Material recycling rates from ICE database demonstrate waste diversion potential. Section N aggregates recycling-weighted diversion rates for construction and demolition waste.
Platinum	Sections A, B, N	Section N provides a live Platinum readiness check: EC reduction $\geq 20\%$ + 100% renewable energy procurement. Status is shown as 'Met' or 'Not Met' with specific gap analysis.

4.2 MRc2: Reduce Embodied Carbon of the Structure and Enclosure

MRc2 (Materials & Resources credit, 2-6 points) requires the project team to conduct a whole-building life-cycle assessment (WBLCA) of the building's structure and enclosure and demonstrate a reduction in embodied carbon compared to a baseline building of equivalent size, function, and location.

- > Baseline EC = 1,050 kgCO₂e/m² GFA (industry average for RC-frame commercial buildings; user-adjustable).
- > Project EC = Sum of A1-A5 emissions per m² GFA, computed from BoQ or parametric model.
- > Reduction (%) = (Baseline - Project) / Baseline x 100.
- > Credit documentation requires: WBLCA report, material quantities, EPDs or ICE factors, and comparison

summary.

4.3 MRc2 Point Thresholds

LEED v5 awards MRc2 points on the following embodied carbon reduction scale:

Reduction	Points	Description
0%	2 pts	Baseline -- complete WBLCA and provide documentation
10%	3 pts	Moderate reduction -- basic material optimisation
20%	4 pts	Significant reduction -- structural efficiency + low-carbon materials
30%	5 pts	Substantial reduction -- advanced material specification
40%+	6 pts	Exemplary performance -- maximum credits

4.4 EAc3: Optimize Energy Performance

EAc3 (Energy & Atmosphere credit, 5-18 points) requires demonstrating energy performance improvement beyond ASHRAE 90.1-2019 baseline through whole-building energy simulation or prescriptive compliance.

- > Baseline EUI = ASHRAE 90.1-2019 reference (varies by building type; default 180 kWh/m²/yr for large offices).
- > Project EUI = Annual operational energy / GFA (from dashboard Section I).
- > B6 Operational Carbon = EUI x GFA x Grid EF, annualised (from dashboard Section B).
- > Scoring starts at 5 points for 5% reduction; +1 point per 1.5% further reduction, capped at 18 points (30.5% reduction).
- > Points are awarded based on energy cost savings or GHG emission reduction, as confirmed by ASHRAE 90.1 Appendix G modelling.

4.5 MRp2: Construction and Demolition Waste Management

MRp2 is a prerequisite (required, no points) that mandates the development and implementation of a construction and demolition waste management plan. The dashboard supports MRp2 compliance by:

- > Tracking end-of-life waste streams (C3-C4) by material type, showing quantities diverted from landfill.
- > Providing recycling rates for each material from the ICE database, supporting waste diversion calculations.
- > The Module D (Section D) credits demonstrate the net benefit of material recovery, which can be cited in the MRp2 waste management plan.

4.6 LEED v5 Platinum Requirements

For LEED v5 Platinum certification (the highest rating level), the project must demonstrate exemplary environmental performance. The dashboard provides a live Platinum readiness check in Section N that verifies the two key carbon-related requirements:

- > Requirement 1: Embodied Carbon Reduction $\geq$ 20% vs baseline (meets MRc2 significant reduction threshold).
- > Requirement 2: 100% Renewable Energy -- either on-site generation, off-site procurement via power purchase agreements (PPA), or certified renewable energy certificates (RECs).

If both criteria are met, the dashboard displays 'Platinum Ready: Yes' with a green indicator. If either criterion is not met, the dashboard shows 'Platinum Ready: No' with specific gap analysis indicating which requirement is failing and by how much.

Note: These are the carbon-related requirements only. LEED v5 Platinum also requires compliance with additional prerequisites and credits across other categories (Location & Transportation, Sustainable Sites, Water Efficiency, Indoor Environmental Quality, and Innovation) which are outside the scope of this carbon calculator.

4.7 LEED Export & Documentation

The LEED v5 tab (Section N) includes an 'Export LEED Summary' button that generates a formatted PDF ready for LEED submission. The export includes:

- > All metric values: EC per m2, EC reduction %, EUI, EUI reduction %.
- > MRc2 and EAc3 point scores with threshold verification.
- > Carbon hotspot analysis: top 3 building elements by embodied carbon.
- > Platinum readiness status with pass/fail for each criterion.
- > 25-year operational carbon projection with grid decarbonisation trajectory.

A separate 'Export Full Report' button generates a comprehensive PDF of the entire dashboard (all sections A-N including charts), suitable for inclusion as a supplementary submission document in LEED v5 credit documentation packages.

5. Data Sources & Embedded Datasets

5.1 ICE Materials Database

The engine embeds carbon factors for 40 common construction materials from the Inventory of Carbon & Energy (ICE) v3.0 (University of Bath, 2019). Each entry includes:

- > Embodied carbon (kgCO₂e/kg or kgCO₂e/m³)
- > Material density (kg/m³)
- > Recycling rate (% recovered at end of life)
- > Landfill emission factor (kgCO₂e/kg if landfilled)
- > Biogenic carbon storage (kgCO₂e/kg) -- for timber/wood products only

5.2 Grid Emission Factors

The engine embeds grid carbon intensity data for 20 countries/regions across 4 decarbonisation scenarios (Conservative, Moderate, Ambitious, Net Zero). Data points for 2020, 2030, 2040, and 2050 with linear interpolation for intermediate years. Sources: IEA World Energy Outlook, national grid operator projections.

5.3 Biogenic Carbon Factors

Timber and wood-based materials store biogenic carbon. The embedded dataset includes biogenic factors for CLT (-0.85 kgCO₂e/kg), glulam (-0.80 kgCO₂e/kg), timber framing (-1.00 kgCO₂e/kg), plywood (-0.78 kgCO₂e/kg), and wood fibre insulation (-0.65 kgCO₂e/kg). Negative values indicate net carbon sequestration.

6. Estimated vs Actual Data -- Dashboard Transparency

The dashboard includes a data source badge on every section, showing whether the underlying data comes from the input JSON ('Actual'), is partially computed ('Partial'), or is fully estimated from available parameters ('Estimated'). This section documents what each badge means for each lifecycle stage.

Stage	Section	Badge	Explanation
A1-A3	Section A	Actual	Data from stages.A1A3 in JSON (material carbon from BoQ or parametric model)

A4	Section A	Actual	Data from stages.A4 or EC_breakdown.A4
A5	Section A	Actual	Data from stages.A5 or EC_breakdown.A5
B1	Section B	Actual	Data from stages.B1 or OC_breakdown.B1_refrigerant
B2	Section A	Estimated*	*Zero in JSON (B_structural=0). No in-use structural repair data provided.
B3	Section A	Estimated*	*Zero in JSON (B_service=0). No in-use service repair data
B4	Section A	Estimated*	*Not in JSON. Replacement data not provided.
B5	Section A	Actual	Data present (B_refurbish=613k kgCO2e).
B6	Section B	Actual	Data from B6_elec + B6_fossil in JSON.
B7	Section B	Estimated	Not in JSON. Estimated from water intensity benchmarks x area x lifetime.
C1	Section C	Estimated	Not in JSON. Estimated at 4.5 kgCO2e/m2 GFA demolition.
C2	Section C	Actual	Data from C_breakdown.C2 in JSON.
C3	Section C	Actual	Data from C_breakdown.C3 in JSON.
C4	Section C	Actual	Data from C_breakdown.C4 in JSON.
D	Section D	Actual	Data from D_breakdown (steel recycling credit).
Offset	Section L	Estimated	Computed from WLC x lifetime using woodland sequestration.
TimeSeries	Section K	Estimated	Generated year-by-year with 2%/yr grid decarbonisation.
Oper.Energy	Section I	Estimated*	*Uses EUI from JSON if available; otherwise CIBSE benchmarks.
3D Model	Section H	Actual	Geometry from floorPlans/floorLevels in JSON.
BoQ	Sections F,G	Actual	Material quantities from boqReport in JSON.

READING THE BADGES

Green 'Actual' = Direct from JSON input, no estimation applied. Purple 'Partial' = Mix of actual and estimated components (e.g., Section E stage summary). Orange 'Estimated' = Fully computed by the engine using benchmarks or extrapolation from available data. All estimates are documented in this guidebook.

7. Variable Building Lifetime Extrapolation

The dashboard includes a lifetime slider (20-120 years) that allows users to explore how building service life affects whole-life carbon. This section explains the extrapolation method.

7.1 How It Works

The engine decomposes carbon into lifetime-independent and lifetime-dependent components:

- > Lifetime-INDEPENDENT (fixed): A1-A5 upfront embodied, C1-C4 end-of-life, D credits. These do not change with building age.
- > Lifetime-DEPENDENT (annual): B6 operational energy, B7 operational water, B1 refrigerant, B2-B5 refurb/repair. These scale linearly with years.

When the input JSON does not contain explicit annual breakdown fields (`_annualOC_kg`, `_annualB6_kg`, etc.), the engine computes them by dividing the total OC by the building lifetime. For example, if the JSON reports OC = 79,858 tCO2e with lifetime = 50 years, the engine derives annual OC = $79,858 / 50 = 1,597$ tCO2e/yr.

7.2 Grid Decarbonisation in Time-Series

Section K (time-series chart) applies a 2%/year linear reduction to the annual operational carbon to model grid decarbonisation. This creates the dashed 'decarbonised' projection line. The decarbonisation rate floors at 20% of the starting value (year 40 onwards).

7.3 Limitations of Lifetime Extrapolation

- > Assumes linear scaling of operational carbon. In reality, equipment efficiency degrades and refurbishment cycles are non-linear.
- > B2-B5 refurbishment is assumed to be distributed evenly per year, which is a simplification.
- > Grid decarbonisation uses a uniform 2%/year assumption regardless of country scenario.
- > EOL and D credits are held constant regardless of lifetime changes (demolition method and recycling rates are assumed independent of building age).

8. Carbon Offset Methodology (Section L)

8.1 Woodland Sequestration Rate

The offset calculation uses a mature woodland sequestration rate of 4.5 tCO₂e per hectare per year, based on UK Woodland Carbon Code data for mixed broadleaf woodland in temperate climates. This is considered conservative for tropical/subtropical regions where growth rates are typically higher.

8.2 Offset Cost Rates

Cost rates applied (in Hong Kong Dollars):

- > Woodland offset (tree planting): HK\$250/tCO₂e (approx. GBP 25/tCO₂e)
- > Estate improvement (building retrofit): HK\$950/tCO₂e (approx. GBP 95/tCO₂e)

GBP to HKD conversion: 1 GBP = 10 HKD (approximate, July 2026).

8.3 Land Area Equivalents (Hong Kong Context)

To make the offset land area requirement more relatable for Hong Kong users, the dashboard provides contextual equivalents:

- > Football field: 105m x 68m = 0.714 hectares (FIFA standard pitch)
- > Victoria Park (Hong Kong): ~19 hectares (the largest public park on Hong Kong Island)
- > Land area in km² is also shown for large projects.

For example, a building with 90,000 tCO₂e WLC over 50 years would require ~400 hectares of woodland to offset, equivalent to approximately 560 football fields or ~21 Victoria Parks.

9. Grasshopper Integration -- Variable Lifetime Data

For the lifetime slider to work optimally with Grasshopper-generated JSON files, the export script should include annualised operational carbon fields. This allows the dashboard to scale operational carbon precisely rather than relying on the dashboard-side extrapolation.

9.1 Recommended JSON Fields from Grasshopper

The following fields should be added to the Grasshopper export script:

Field	Type	Description
_baseLifetime	number	The reference lifetime in years (e.g., 50).
_annualOC_kg	kgCO2e	Total annual operational carbon = (B6+B1)/lifetime.
_annualB1_kg	kgCO2e	Annual B1 refrigerant leakage = B1/lifetime.
_annualB6_kg	kgCO2e	Annual B6 energy carbon = (B6_elec+B6_fossil)/lifetime.
_annualB7_kg	kgCO2e	Annual B7 water carbon = B7/lifetime.
_fixedEC_kg	kgCO2e	Lifetime-independent EC = (A1-A5 + B2-B5 fixed) x 1000.
_fixedEOL_kg	kgCO2e	Lifetime-independent EOL = (C1-C4) x 1000.
_fixedD_kg	kgCO2e	Lifetime-independent D credits = D x 1000.

9.2 Grasshopper Python Component Example

In a Grasshopper Python component, add the following to your JSON output:

```
import json

# ... existing calculations ...

lifetime = 50 # building design life

annual_oc = (b6_elec + b6_fossil + b1) / lifetime

output_json['_baseLifetime'] = lifetime
output_json['_annualOC_kg'] = annual_oc
output_json['_annualB1_kg'] = b1 / lifetime
output_json['_annualB6_kg'] = (b6_elec + b6_fossil) / lifetime
output_json['_fixedEC_kg'] = ec_total * 1000 # A1-A5, in kg
output_json['_fixedEOL_kg'] = (c1 + c2 + c3 + c4) * 1000
output_json['_fixedD_kg'] = d_total * 1000
```

With these fields present, the dashboard slider will show immediately upon loading, and all chart data will be recomputed client-side with accurate annual decomposition rather than the uniform extrapolation fallback.

10. Limitations & Caveats

10.1 Parametric BoQ Accuracy

When no actual BoQ is provided, the parametric estimation uses simplified assumptions about structural design (column/beam spacing, slab thickness, reinforcement ratios). These produce reasonable first-order estimates (+/-30% typical accuracy) but cannot replace a detailed BoQ for design-stage or as-built carbon accounting.

10.2 Regional Adaptation

The ICE database is based on UK/European manufacturing data. Embodied carbon factors may differ in other regions due to different manufacturing energy mixes, transport distances, and construction practices. The grid EF data partially

addresses this for operational carbon.

10.3 Biogenic Carbon Timing

Biogenic carbon storage in timber is treated as an instantaneous credit at year 0, which is consistent with EN 15804 but does not reflect the time-value of carbon (e.g., PAS 2080 or dynamic LCA approaches). The sequestered carbon is assumed to remain stored for the building's lifetime and is released at end-of-life (C4).

10.4 Water Carbon (B7)

The water carbon estimation uses UK benchmarks (CIBSE/BSRIA) and may not accurately reflect Hong Kong's water supply carbon intensity (which involves energy-intensive desalination and long-distance transfer from Dongjiang). Users should replace default values with project-specific data where available.

10.5 What the Dashboard Does NOT Do

- > It does NOT perform dynamic thermal simulation or operational energy modelling.
- > It does NOT account for future climate change impacts on heating/cooling demand.
- > It does NOT perform Monte Carlo uncertainty analysis on parametric estimates.
- > It does NOT calculate upfront biogenic carbon separately from operational biogenic carbon.
- > It does NOT distinguish between Scope 1, 2, and 3 emissions (all are lifecycle-based per EN 15978).
- > It does NOT calculate non-CO2 environmental impacts (e.g., acidification, eutrophication, ozone depletion).

11. Glossary

BoQ	Bill of Quantities -- a detailed list of materials, parts, and labour for a construction project.
CLT	Cross-Laminated Timber -- an engineered wood product used for structural walls, floors, and roofs.
EC	Embodied Carbon -- total carbon emissions from materials and construction (A1-A5 + B2-B5).
EF	Emission Factor -- the carbon intensity of an energy source, measured in gCO ₂ e/kWh.
EOL	End of Life -- the final lifecycle stage including demolition, transport, waste processing, and disposal (C1-C4).
EUI	Energy Use Intensity -- annual energy consumption per unit floor area (kWh/m ² /yr).
GFA	Gross Floor Area -- the total floor area of a building measured to the external walls (m ²).
GWP	Global Warming Potential -- a measure of how much heat a greenhouse gas traps relative to CO ₂ over 100 years.
ICE	Inventory of Carbon & Energy -- a database of embodied carbon factors for construction materials (University of Bath).
OC	Operational Carbon -- carbon emissions from building energy and water use during operation (B1, B6-B7).
RC	Reinforced Concrete -- concrete with steel reinforcement bars.
tCO₂e	Tonnes of Carbon Dioxide Equivalent -- the standard unit for carbon accounting.
WLC	Whole-Life Carbon -- the sum of EC + OC + EOL + D (all lifecycle stages).
WWR	Window-to-Wall Ratio -- the proportion of external wall area that is glazed.